

MEDICAL EQUIPMENT

Standards & Interoperability Study

HORIBA ABX Micros 60

Abaxis Piccolo Xpress

Analysis of FHIR R4, HL7, and LOINC Support

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Context: Reference study for Smartband V001 FHIR integration

Scope: HL7 messaging support · FHIR R4 compatibility · LOINC coding · LIS connectivity · Smartband V001 relevance

DEVICE OVERVIEW — At a Glance

HORIBA ABX Micros 60

Type	Hematology Analyzer (CBC)
Manufacturer	HORIBA ABX SAS (France)
Throughput	60 samples/hour (open tube)
Sample Vol.	10 µL whole blood
Method	Impedance + Photometry (HGB)
Parameters	18 CBC parameters
HL7	Via LiteDM middleware bridge
Connectivity	RS-232 serial output
FHIR	Indirect via middleware

Abaxis Piccolo Xpress

Type	Point-of-Care Chemistry Analyzer
Manufacturer	Abaxis (now Abbott Point of Care)
Test Time	12 minutes per panel
Sample Vol.	100 µL (whole blood/serum/plasma)
Method	Spectrophotometry (9 wavelengths)
Parameters	31 chemistry analytes / 17 panels
HL7	Native HL7 v2.x output
Connectivity	5× USB + 1× Ethernet
FHIR	Indirect via middleware

ABX MICROS 60 — 18 CBC Parameters with LOINC Codes

Abbrev.	LOINC	Unit	Parameter
WBC	6690-2	10 ³ /μL	White Blood Cell Count
RBC	789-8	10 ⁶ /μL	Red Blood Cell Count
HGB	718-7	g/dL	Hemoglobin
HCT	4544-3	%	Hematocrit
MCV	787-2	fL	Mean Corpuscular Volume
MCH	785-6	pg	Mean Corp. Hemoglobin
MCHC	786-4	g/dL	Mean Corp. Hgb Concentration
RDW	788-0	%	Red Cell Distribution Width
PLT	777-3	10 ³ /μL	Platelet Count

Abbrev.	LOINC	Unit	Parameter
MPV	32623-1	fL	Mean Platelet Volume
PCT	51637-7	%	Platelet Crit
PDW	32207-3	%	Platelet Distribution Width
LYM%	736-9	%	Lymphocyte %
LYM#	731-0	10 ³ /μL	Lymphocyte #
MON%	5905-5	%	Monocyte %
MON#	742-7	10 ³ /μL	Monocyte #
GRA%	764-1	%	Granulocyte %
GRA#	30180-4	10 ³ /μL	Granulocyte #

PICCOLO XPRESS — 31 Chemistry Analytes with LOINC Codes

Abbr.	LOINC	Unit	Analyte	Abbr.	LOINC	Unit	Analyte	Abbr.	LOINC	Unit	Analyte
ALB	1751-7	g/dL	Albumin	CRE	2160-0	mg/dL	Creatinine	tCO ₂	2028-9	mmol/L	Total Carbon Dioxide
ALP	6768-6	U/L	Alkaline Phosphatase	CRP	1988-5	mg/L	C-Reactive Protein	TP	2885-2	g/dL	Total Protein
ALT	1742-6	U/L	Alanine Aminotransferase	GGT	2324-2	U/L	Gamma-Glutamyl Transferase	UA	3084-1	mg/dL	Uric Acid
AMY	1798-8	U/L	Amylase	GLU	2345-7	mg/dL	Glucose	CHOL	2093-3	mg/dL	Total Cholesterol
AST	1920-8	U/L	Aspartate Aminotransferase	K ⁺	2823-3	mmol/L	Potassium	HDL	2085-9	mg/dL	HDL Cholesterol
DBIL	1968-7	mg/dL	Direct Bilirubin	LAC	2524-7	mmol/L	Lactate	LDL*	13457-7	mg/dL	LDL Cholesterol (calc.)
TBIL	1975-2	mg/dL	Total Bilirubin	LDH	2532-0	U/L	Lactate Dehydrogenase	nHDL*	70204-3	mg/dL	Non-HDL Cholesterol (calc.)
BUN	3094-0	mg/dL	Blood Urea Nitrogen	Mg	2601-3	mg/dL	Magnesium	TRIG	2571-8	mg/dL	Triglycerides
Ca	17861-6	mg/dL	Total Calcium	Na ⁺	2951-2	mmol/L	Sodium	VLDL*	13458-5	mg/dL	VLDL Cholesterol (calc.)
CK	2157-6	U/L	Creatine Kinase	PHOS	2777-1	mg/dL	Phosphorus	CHOL/HDL*	9830-1	ratio	Cholesterol/HDL Ratio (calc.)
Cl ⁻	2075-0	mmol/L	Chloride								

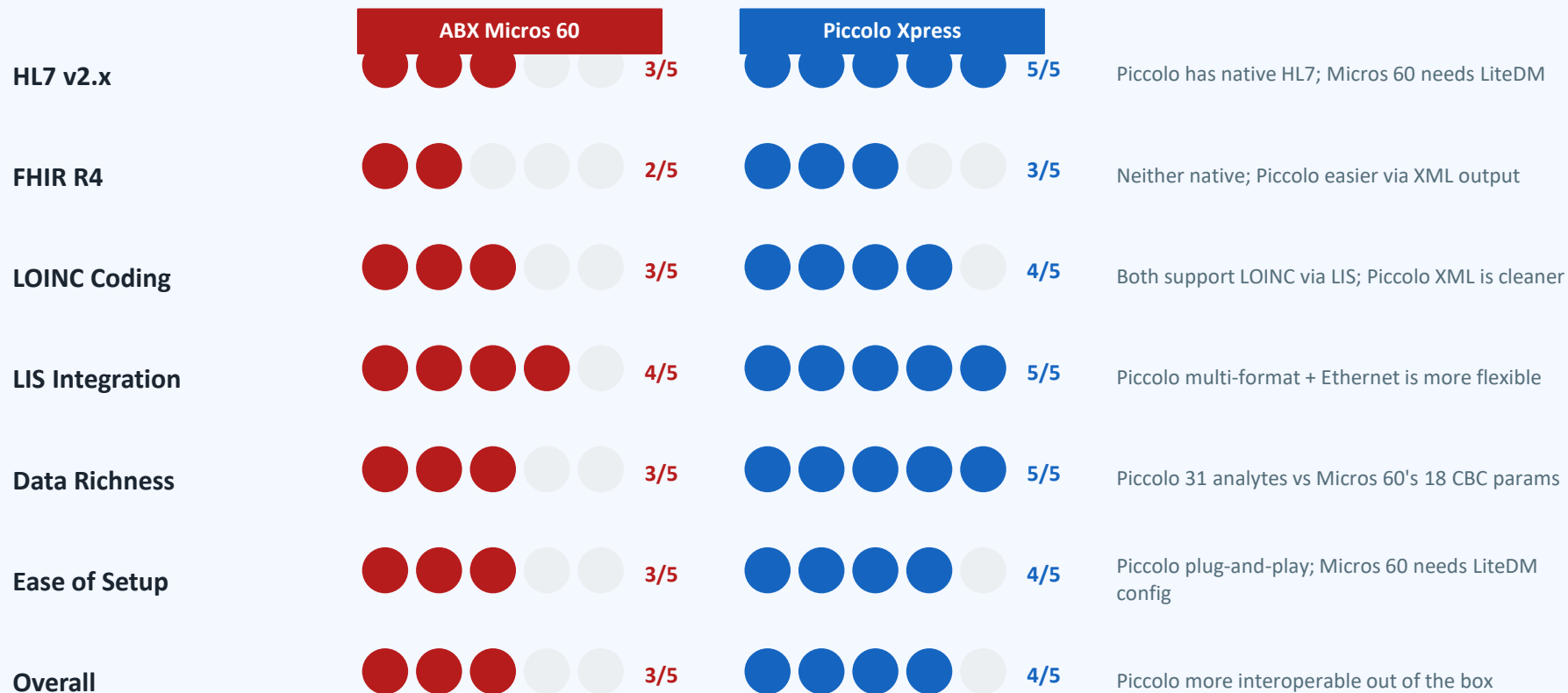
HL7 & FHIR SUPPORT COMPARISON

HL7 Support:

Feature	ABX Micros 60	Piccolo Xpress
Native HL7 Output	No — requires LiteDM middleware	Yes — native HL7 v2.x output
HL7 Version	HL7 v2.x via LiteDM bridge	HL7 v2.x (ASCII, ASTM, HL7, XML)
Message Direction	Bidirectional via LiteDM	Bidirectional (USB + Ethernet)
Physical Interface	RS-232 serial	5× USB + 1× Ethernet
EMR Compatibility	Athena, Flatiron, Epic, eClinicalWorks, CureMD	All HL7-compatible EMR/LIS systems
Setup Complexity	Requires LiteDM install + HL7 bridge config	Plug-and-play with HL7-capable systems

FHIR R4: Neither device supports it natively. Both require middleware (Mirth Connect / Relaymed) to transform HL7 v2.x → FHIR R4 DiagnosticReport + Observation resources.

STANDARDS READINESS SCORES



PICCOLO XPRESS — 17 Panel Configurations

Panel Name	Cat. #	Tests Included	CLIA
General Chemistry 6	400-0006	ALT, AST, BUN, CRE, GGT, GLU	●
General Chemistry 13	400-0029	ALB, ALP, ALT, AMY, AST, BUN, Ca, CRE, GGT, GLU, TBIL, TP, UA	●
Basic Metabolic Panel	400-0024	BUN, Ca, Cl ⁻ , CRE, GLU, K ⁺ , Na ⁺ , tCO ₂	●
Comprehensive Metabolic	400-0028	ALB, ALP, ALT, AST, BUN, Ca, Cl ⁻ , CRE, GLU, K ⁺ , Na ⁺ , TBIL, tCO ₂ , TP	●
Electrolyte Panel	400-0022	Cl ⁻ , K ⁺ , Na ⁺ , tCO ₂	●
Lipid Panel	400-0025	CHOL, CHOL/HDL*, HDL, nHDL*, LDL*, TRIG, VLDL*	●
Lipid Panel Plus	400-0030	ALT, AST, GLU + full lipid panel	●
Hepatic Function Panel	400-0026	ALB, ALP, ALT, AST, DBIL, TBIL, TP	●
Liver Panel Plus	400-0003	ALB, ALP, ALT, AMY, AST, GGT, TBIL, TP	●
Renal Function Panel	400-0027	ALB, BUN, Ca, Cl ⁻ , CRE, GLU, K ⁺ , Na ⁺ , PHOS, tCO ₂	●
MetLyte 8	400-0023	BUN, CK, Cl ⁻ , CRE, GLU, K ⁺ , Na ⁺ , tCO ₂	●
MetLyte Plus CRP	400-0034	BUN, CK, Cl ⁻ , CRE, CRP, GLU, K ⁺ , Na ⁺ , tCO ₂	●
MetLac 12	400-0037	ALB, BUN, Ca, Cl ⁻ , CRE, GLU, K ⁺ , LAC, Mg, Na ⁺ , PHOS, tCO ₂	●
BMP Plus	400-0031	BUN, Ca, Cl ⁻ , CRE, GLU, K ⁺ , LDH, Mg, Na ⁺ , tCO ₂	●
BioChemistry Panel Plus	400-0035	ALB, ALP, ALT, AMY, AST, BUN, Ca, CRE, CRP, GGT, GLU, TP, UA	●
AmLyte 13	400-0041	ALB, ALT, AMY, AST, BUN, Ca, CK, CRE, CRP, GLU, K ⁺ , Na ⁺ , TBIL	●
Kidney Check CLIA Waived	400-0033 11 of 17 panels are CLIA waived	BUN, CRE	●

● Not waived

RELEVANCE TO SMARTBAND V001 & FHIR INTEGRATION

LOINC Code Reference for Smartband V001:

Smartband Measurement	Analyzer Source	LOINC	Connection
Heart Rate (MAX30102)	Not in these analyzers	8867-4	Shared LOINC coding approach
SpO2 (MAX30102)	Not in these analyzers	2708-6	Shared LOINC coding approach
Glucose (future sensor)	Piccolo Xpress: GLU	2345-7	Direct LOINC reuse from Piccolo
CRP (inflammation)	Piccolo Xpress: CRP	1988-5	Direct LOINC reuse from Piccolo
Hemoglobin (derived)	ABX Micros 60: HGB	718-7	Reference for wearable Hgb estimation
Electrolytes (future)	Piccolo Xpress: Na ⁺ , K ⁺ , Cl ⁻	Multiple	Reference for future electrolyte sensor

Key Learnings for Smartband V001 FHIR Implementation:

- Both analyzers use HL7 v2.x ORU^R01 messages — same type that would carry Smartband V001 data via a gateway
- Neither generates FHIR R4 natively — confirming middleware-based FHIR transformation is the industry standard
- LOINC codes assigned by LIS/middleware layer, not by the device — consistent with Smartband V001's FHIR gateway approach
- DiagnosticReport + Observation FHIR pattern (used by both) is the same pattern Smartband V001 FHIR server implements

CONCLUSION

ABX Micros 60 ★★★ (3/5)

- Older RS-232 based design
- HL7 requires LiteDM middleware
- 18 CBC parameters — all LOINC coded
- FHIR via Mirth Connect middleware
- Reliable but requires extra config

Piccolo Xpress ★★★★★ (4/5)

- Modern USB + Ethernet connectivity
- Native HL7, ASTM, XML, ASCII output
- 31 analytes / 17 panels — all LOINC coded
- FHIR via Relaymed or middleware
- Plug-and-play — more interoperable

Neither device natively supports FHIR R4 — confirming that middleware-based FHIR transformation (as in Smartband V001's Flask server) is the correct industry-standard approach.